

Problem 7.2

Solution:**Construct the Bipartite Graph:**

1. On the left side, create one node e for each edge $e \in E$ of the original graph G .
2. On the right side, create one node v for each vertex $v \in V$ of the original graph G .
3. Add a source node s and a sink node t .
4. For each edge node e on the left, add an arc $(s \rightarrow e)$ of capacity 1, representing the need to pick exactly one direction for that edge.
5. Suppose the edge e in G connects vertices u and v , add arcs $(e \rightarrow u)$ and $(e \rightarrow v)$, each with capacity 1. This enforces that the 1 unit of flow from edge e can be sent either from vertex u or from vertex v , but not both simultaneously (by the integrality theorem).
6. For each vertex v on the right side, add an arc $(v \rightarrow t)$ of capacity d . The capacity d enforces that at most d edges can flow **out** of vertex v .
7. Compute the maximum flow using Ford-Fulkerson.
8. If there is a flow of value $|E|$, that means we managed to push 1 unit of flow for every edge node to exactly one of its two endpoints, without exceeding the capacity d at any vertex. Hence we get a valid orientation with out-degree $\leq d$.
9. If the max flow is less than $|E|$, then no assignment of directions can keep every vertex's out-degree $\leq d$.

Proof of Correctness:

1. If there is an orientation of G with out-degree $\leq d$, we can construct a flow of value $|E|$:
 - For each edge $e = \{u, v\}$, suppose it is oriented $u \rightarrow v$. Then send 1 unit of flow from s to edge node e , and then from e to the vertex u in the flow network.
 - Because each vertex v has at most d edges oriented out of it in the original orientation, it receives flow from at most d of its edges. Hence the flow into v is at most d , so the capacity $(v \rightarrow t)$ is never exceeded.
 - Thus, we achieve an integer flow of size exactly $|E|$.
2. If the max flow in the constructed network is $|E|$, then each edge node e sends exactly 1 unit of flow to exactly one of its endpoints in G . That determines a direction of e . Since no vertex node v can exceed capacity d to the sink, at most d edges are directed out of any vertex v .

Hence the flow network construction is both sound (no invalid orientations can achieve full flow) and complete (every valid orientation induces a full flow).